Zack Murry

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Education

University of Missouri — August 2023 – May 2027
B.S. Computer Science, B.S. Mathematics — GPA: 4.0/4.0

Experience

Capital One — June – August 2026
Software Engineering Intern — *New York, NY*

- Built a Node.js backend service on AWS Lambda to aggregate account data from upstream APIs into tailored, product-specific responses for onboarding 160k+ users monthly to bank accounts
- Integrated the service into the Angular web client using NgRx state management, building reactive pipelines to handle asynchronous data flows and recover gracefully from upstream API failures
- Designed an OpenAPI contract and unified Zod schema for request validation and automated API documentation
- Architected a normalized data schema that consolidates shared content into a single source with per-product overrides, cutting redundant updates by over 90%

University of Missouri College of Engineering — August 2023 – May 2026
Undergraduate Research Assistant — *Columbia, MO*

- Created a containerized simulation orchestration framework for modeling drone flight behavior with PX4 and Docker, enabling reproducible validation of routing algorithms across 300+ simulations with a 95% success rate
- Built a full-stack Next.js web app for planning and optimizing truck-drone package delivery routes with an interactive map and Python-based combinatorial optimization
- Co-authored 3 peer-reviewed papers at IEEE INFOCOM and ICNC on drones, simulation, and edge systems

Garmin — May – August 2025
Software Engineering Intern — *Olathe, KS*

- Engineered Java Spring microservice endpoints for a new nutrition tracking feature in the Garmin Connect API, supporting the launch of calorie and macronutrient tracking across the mobile app and smartwatches
- Built a batch-processing service for high-throughput writes to HBase with Protobuf, decreasing processing time by 63% using a custom data structure to track changes in nutrition settings and goals over time
- Implemented device gateway APIs with Spring Cloud Hystrix for fault-tolerant smartwatch-to-server communication, reducing payload size by 40% to improve performance on memory-constrained devices

University of Chicago — May – August 2024
Software Research Intern — *Chicago, IL*

- Designed and deployed a Hadoop cluster on 6 Raspberry Pis to benchmark distributed computing across rural 5G radio links, building custom ARM64 Alpine Linux Docker images and automating deployment with K3s
- Soldered UART GPS modules (GT-U7) to each node and configured NTP synchronization using chrony and PPS (pulse-per-second) signals, achieving sub-microsecond clock error across the cluster
- Built a real-time telemetry and observability pipeline using MQTT, InfluxDB, and Grafana to monitor network throughput and cluster health across distributed edge nodes

Projects

Docs Hotkey | *TypeScript, React* — `docs-hotkey.zackmurry.com`

- Developed an open-source Chrome extension for adding custom keyboard shortcuts to Google Docs
- Reverse-engineered dynamic DOM and event handling to implement 15+ configurable hotkey actions
- Grew the extension to 12,000+ weekly active users and a 4.6-star rating with over 90 reviews

ChesSRS | *Kotlin, Spring, React, PostgreSQL, Docker, Redis* — `chessrs.zackmurry.com`

- Designed a spaced-repetition system (SRS) for studying chess openings using an interactive chessboard
- Integrated the Lichess API for OAuth2 authentication, game-history imports, and access to 100K+ studies
- Built Spring Boot GraphQL APIs backed by PostgreSQL to manage study schedules and chessboard interaction

Technical Skills

Languages: Java, Python, C++, TypeScript, Kotlin, JavaScript, Bash
Tools: Spring Boot, Node.js, React, Next.js, AWS, Docker, Kubernetes, Linux, PostgreSQL, HBase, Redis, Git